

NHS OUTPATIENT PERFORMANCE ANALYSIS REPORT

A Data-Driven Study of Capacity, Efficiency & Telehealth Opportunity

113.2M	~8.9M	17.99%	87–114K
Total Appointments	Usable Diagnosis Codes	Current Telehealth Adoption	Potential Capacity Opportunity

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Portfolio Project — Business & Data Analytics

1. Executive Summary

This project analysed **113.2 million** NHS outpatient appointments to identify strategic opportunities for capacity expansion, operational efficiency, and telehealth adoption. The analysis combines Python-based data processing with Power BI visualisation to surface actionable insights for service planning and digital transformation teams.

Headline Finding — Data Quality

The single most significant finding is a systemic data completeness issue: **92.1%** of all outpatient attendances (104.3 million) are recorded under “*Unknown and unspecified causes of morbidity*” rather than a specific ICD-10 diagnosis. Only **7.9%** (~8.9 million) carry a usable diagnosis code. This means any diagnosis-level analysis is currently built on a minority of total activity. Closing this gap should be treated as the top-priority recommendation.

Three Strategic Opportunities

Opportunity	Focus	Estimated Impact
1. Diagnosis Coding	Raise usable coding from 7.9% toward 95%	Enables reliable pathway analysis; foundation for all other improvements
2. Follow-up Optimisation	Reduce extreme variation (0.7–14.1 follow-ups per patient)	22K–44K appointment slots potentially releasable
3. Telehealth Expansion	Move from 17.99% toward clinically suitable potential (~35–45%)	50K+ additional remote capacity opportunity

Combined capacity opportunity (illustrative): 87,000–114,000 additional appointments (approximately 6.6–8.7% of current volume). These figures are scenario-based and require clinical validation.

2. Background & Business Questions

Outpatient services account for the largest share of NHS elective activity. With demand continuing to outstrip capacity, service planning teams need an evidence-based view of where demand is concentrated, which conditions generate the heaviest repeat-attendance burden, and where telemedicine is under-used relative to its potential.

This analysis addresses four core questions:

- Where is outpatient demand concentrated, and is that concentration in conditions suited to standardisation or digital pathways?
- Which high-volume diagnoses generate disproportionate follow-up burden?
- Which high-volume diagnoses are least digitally adopted, representing clear telemedicine investment opportunities?
- Is the underlying data fit for this kind of decision-making?

3. Headline Finding: Diagnosis Coding Completeness

Of **113,225,153** total outpatient attendances, **104,324,890 (92.1%)** are recorded against “Unknown and unspecified causes of morbidity” rather than a specific ICD-10 diagnosis. Only **8,900,245 attendances (7.9%)** carry a specific, analytically useful diagnosis code.

This is materially different from normal background data noise. At this scale it indicates a systemic gap in diagnosis capture at the point of care — most likely concentrated in particular specialties, systems, or recording workflows. Until coding completeness improves substantially, diagnosis-level workforce, capacity, and digital-adoption analysis will remain limited in reliability.

Metric	Value
Total outpatient attendances	113,225,153

Attendances with NO usable diagnosis	104,324,890 (92.1%)
Attendances with usable diagnosis code	8,900,245 (7.9%)
Diagnoses driving ~80% of coded demand	568 of 7,129 (~8%)

4. Demand Concentration

Among the coded subset, demand is highly concentrated. The top diagnoses by attendance volume are:

Rank	Primary Diagnosis	Attendances
1	Malignant neoplasm of prostate	213,533
2	Pregnant state, incidental	193,739
3	Malignant neoplasm: Breast, unspecified	178,154
4	After-cataract	171,334
5	Pain in joint	166,330
6	Degeneration of macula and posterior pole	156,386
7	Other specified general symptoms and signs	153,254
8	Other senile cataract	127,368
9	Examination of eyes and vision	93,506
10	Supervision of normal pregnancy, unspecified	83,704

The top five diagnosis groups account for a large share of coded outpatient demand. Oncology, maternity-related, and ophthalmology pathways dominate volume and should be prioritised for pathway review.

5. Operational Efficiency — Follow-up Burden

Across coded diagnoses, the average follow-up ratio is approximately **1.6** subsequent appointments per first appointment, but the range is extreme: **0.70 to 14.10**. High ratios may reflect appropriate chronic disease monitoring (e.g. macular degeneration, oncology) or, in some cases, lack of clear discharge criteria and defensive practice.

Diagnosis	Attendances	Follow-up Ratio	Tele Adoption
Degeneration of macula and posterior pole	156,386	11.40	6%
Malignant neoplasm of prostate	213,533	11.10	25%
Malignant neoplasm: Breast, unspecified	178,154	7.40	21%
Pregnant state, incidental	193,739	4.70	7%
Pain in joint	166,330	1.60	13%
After-cataract	171,334	1.30	0%

Diagnoses with both high volume and high follow-up ratios represent the greatest efficiency opportunity. Pathway standardisation, clearer discharge criteria, and selective use of patient-initiated follow-up (PIFU) or remote monitoring could release significant capacity. Illustrative modelling suggests **22,000–44,000** slots could be optimised, subject to clinical validation.

6. Telehealth Opportunity

Current telehealth adoption stands at **17.99%** (~2 million remote attendances). Several low-volume diagnoses already achieve 100% remote delivery (e.g. certain infectious and respiratory conditions), proving feasibility for low-acuity pathways. However, many high-volume diagnoses remain near zero remote adoption despite clinical suitability for hybrid or fully remote models.

A diagnosis-level suitability assessment (exam requirements, acuity, monitoring capability, patient-reported symptoms) suggests that overall remote potential lies in the region of **35–45%**. Expanding telehealth for high-volume, clinically appropriate pathways could unlock an estimated **50,000+** additional remote capacity opportunities while protecting in-person slots for complex cases.

Priority starting points include high-volume diagnoses currently at or near 0% remote adoption (e.g. after-cataract, certain ophthalmology reviews) and stable chronic pathways that already show moderate adoption but have room to grow.

7. Recommendations

1. Treat diagnosis coding completeness as the top priority

Without improving the 7.9% usable coding rate, all diagnosis-level planning remains limited. Recommended actions: root-cause analysis of missing codes, pre-visit questionnaires, clinician training, and system prompts at the point of care.

2. Pilot pathway standardisation and PIFU on high-burden diagnoses

Focus on macular degeneration review, cataract follow-up, and selected oncology pathways. Establish evidence-based ideal follow-up intervals and clear discharge criteria.

3. Target telemedicine expansion at high-volume, low-adoption diagnoses

Avoid blanket “move everything to video” policies. Start with diagnoses that already show clinical suitability and near-zero current adoption. Validate safety with clinical leads before scaling.

4. Re-run the analysis quarterly

Convert this static snapshot into a trend view, tracking coding completeness and telehealth adoption over time using NHS England published outpatient activity releases.

8. Data, Method & Limitations

Source: National snapshot of outpatient attendances by primary diagnosis (ICD-10 coded). No time dimension or trust-level breakdown was available in the analysed extract.

Method notes:

- “Unknown and unspecified causes of morbidity” and total activity rows were excluded from diagnosis-level rankings but reported separately as the headline data-quality finding.
- Follow-up and telehealth rankings were restricted to higher-volume diagnoses to avoid statistically extreme but meaningless ratios from very low-volume codes.
- Priority scoring used percentile ranks with transparent weighting so methodology can be challenged and re-weighted by stakeholders.

Limitations:

- Analysis is retrospective and based on a single snapshot.
- No geographic or trust-level variation is available.
- Capacity and telehealth opportunity figures are illustrative scenarios, not forecasts. They assume trends continue and require clinical validation before use as planning targets.
- Actual telehealth adoption depends on clinical appropriateness, patient preference, technology, and workforce readiness.

9. Tools & Approach

- **Python** (pandas, numpy, matplotlib, seaborn) — data processing, quality assessment, scoring models
- **Power BI** — interactive dashboards (Executive Summary, Demand Analysis, Operational Efficiency, Tele-Health Opportunity)

- Techniques: data quality profiling, ratio analysis, diagnosis suitability scoring, scenario modelling

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This report is intended for portfolio demonstration purposes. Findings are illustrative and require clinical and operational validation.